Constructed factors at every retained crossing node

14 November 2026; 42 authored Lean statements
Source `708762cbcabfba2c185ae8d19c53e539901b1369`

Actual scope. The same completed auxiliary orientation polynomial has graded crossing-entry bounds at every retained nonzero source node, explicit first-jet precision factors, and polynomial lowest-layer factors. Local build exits 0 without warnings. The pinned shared audit covers 44 declarations with zero extra trusted axioms. Import-inclusive fresh replay exits 0 with exact 39-byte stdout. All records are unsigned [pcc:unverified].

Node and coefficient conventions. Let $L = 2^a$ and $S = \{-L, \dots, L-1\} \subset \mathbb{Q}$. Take an actual nonzero integer source node $t \in S$ with $v_2(t) = i < u \le a$, and actual ports $x, y < L$ with equal bit i. Use the same completed $W_{xy}(T)$ and natural retained multiplicities m_j. Define

$$G_i = m_i - \sum_{j<u} (j-i)_+ m_j, \quad P_{xy}(Z) = W_{xy}(t + 2^{a+1}Z).$$

The actual nonzero reference c remains the first diagonal jet at the *negative* pole -2^i, with $w = v_2(c) = a+1-i+G_i$. It is not the potentially zero first coefficient at t.

For a polynomial define

$$\mathrm{Graded}(k, P) \iff \forall n \ge 0 : \; v_2([Z^n]P) \ge k + n,$$

with a zero coefficient admitted at every weight. Finite sums, products, powers and scalar polynomials have explicit closure lemmas. In particular Z has grade -1 [pcc:unverified].

Direct bounds that permit zeros. The actual affine anchor factor has grade

$$\left(\begin{cases} i+1 & j = i, \\ \min(i,j) & j \ne i \end{cases} \right) - v_2(2^j - \delta), \qquad \delta = y - x.$$

The constant coefficient follows from the integer numerator divisibility; the coefficient of Z has the additional degree weight. The strict $i < a$ ensures this slope bound even at $j = i$. No division by an anchor numerator or its local value occurs. Derived selected-denominator valuations and the actual bit mask give correction grade at least G_i [pcc:unverified].

The actual crossing bits exclude $\delta = t$ and imply $v_2(t - \delta) \le i$. The literal signed-grid bridge then gives base grade $a - i$, so

$$\mathrm{Graded}(w-1, P_{xy}), \qquad \mathrm{Graded}(-1, P_{xy}/c).$$

Thus normalized coefficient $n+1$ has weight at least n. The actual nodal value makes its constant zero [pcc:unverified].

Actual row comparison and constructed matrix factors

Let $i < u \le v \le a$ and $x \equiv z \pmod{2^v}$. Their actual retained masks agree. Write F_{xy} for the actual masked correction. For an arbitrary value T, including an anchor zero, the literal product identity is

$$F_{xy}(T) = F_{zy}(T) \prod_{j \in M_{xy}} \left(\frac{2^j - \delta(z,y)}{2^j - \delta(x,y)} \right)^{m_j}.$$

Only the actual nonzero selected denominators are cancelled. At the source node t, multiply this ratio by

$$\frac{t - \delta(z,y)}{t - \delta(x,y)}$$

to obtain $Q_t(x,z,y)$. Actual congruence and the derived denominator bounds give $v_2(Q_t - 1) \ge v - u + 1$. Writing $J_{xy}(t) = [Z]P_{xy}$,

$$J_{xy}(t)/c = (J_{zy}(t)/c)Q_t, \quad v_2\big(J_{xy}(t)/c - J_{zy}(t)/c\big) \ge v - u + 1.$$

The prototype's integrality follows from the graded entry theorem. These statements do not assume positive multiplicities or a nonzero local jet [pcc:unverified].

Literal factors. Let R_t be $J_{xy}(t)/c$ on the crossing block and zero elsewhere. For $r \in \mathrm{Fin}(2^v)$, use the actual port with value r, justified by $v \le a$, and set

$$A_{xr} = [x \bmod 2^v = r], \qquad B_{ry} = R_t(r,y).$$

The finite matrix product selects exactly that representative. Both factors are integral and

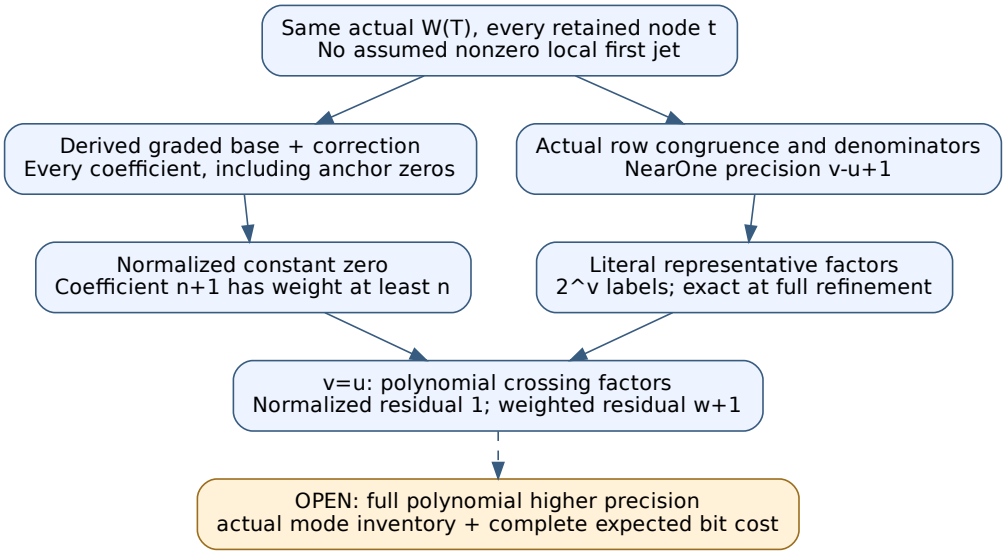
$$v_2\big((R_t - AB)_{xy}\big) \ge v - u + 1.$$

All 2^v labels are charged. At $v = a$ this product is exactly R_t, providing a full-row control, not a cheaper factorization claim [pcc:unverified].

Whole polynomial lowest layer. At $v = u$, normalized entries obey $P_{xy}/c \equiv (J_{xy}(t)/c)Z \pmod 2$ coefficientwise. Set $\widetilde{A} = C(A)$ and $\widetilde{B} = C(B)Z$, using the same 2^u labels. Then the normalized polynomial crossing matrix has error of content at least one from $\widetilde{A}\widetilde{B}$. Scaling the left by the actual $C(c)$ gives row content w and absolute residual $w + 1$. The scaled left is integral when $w \ge 0$, through the unchanged parent lemma. The matrices explicitly zero noncrossing entries; no rank assertion about all of W follows [pcc:unverified].

Anchor zeros and the remaining cost interface

Symbolic zero case. At a positive retained anchor $t = 2^i$, the actual crossing mask contains i. If $m_i > 0$, its correction value is zero. The base value at this other source node is also zero, so the product rule gives $J_{xy}(2^i) = 0$. The all-node bounds retain this case; they do not assume a decomposition $P = JZU$ with a nonzero local J. Positivity is required here, not for the general factor construction [pcc:unverified].



Costs not supplied by a factor count. The reviewed primary proof uses $D_i = L/2^i$, at most $3M/L$ packets, and at most $6M/D_i$ modes per weight. The residue construction retains every 2^u class or 2^v refinement label. This is an extension of node and hypothesis scope, not an original mode-density improvement or an empirical speed result [pcc:unverified].

The polynomial result is the lowest precision layer of every required crossing chart. The refinement precision statement concerns first jets. It does not supply arbitrary-precision factors for the whole polynomial. Next: construct those actual higher-chart moments and price all coefficient, entry, bit, degree, precision, recovery and recursion work. The actual elliptic relation map, public finder, field/scalar extraction, retries and complete expected bit cost remain separate proof obligations.

Proof map, reproducibility and validation status

Current checks. The isolated project builds 42 authored statements without warnings. The pinned shared audit exits 0: 44 declarations, zero extra trusted axioms. Import-inclusive fresh replay exits 0 with exact 39-byte stdout. The completed parent component remains 57 authored statements, gate59/0 and fresh exit0; its source, claims, report/checkpoint/final hashes are preserved [pcc:unverified].

Central source bindings. Full statements and all 42 new unsigned claim records are in `../evidence/claim-pins.json`.

`actual_node_coefficient_weight`	PCC-20261114-75d18d
`positive_anchor_first_zero`	PCC-20261114-c74b5e
`actual_node_precision_comparison`	PCC-20261114-538559
`actual_node_precision_factorization`	PCC-20261114-b2f99f
`actual_node_chart_factorization`	PCC-20261114-64b8cc
`actual_node_weighted_chart_factorization`	PCC-20261114-28bc8e

The complete Lean statement governs each hypothesis. There is no supplied good content, integral basis or rank witness [pcc:unverified].

Pins. Lean 4.35.0, CryptoLean and Mathlib revisions:

`148f5dc389b341401e06d76abe7783c4ccf42f2e`
`cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`

Source, manifest, gate, audit and trust bytes are listed in `../evidence/source-freeze.json`. Only the standard foundations `propext`, `Classical.choice`, `Quot.sound` appear in printed central dependencies. Gate data is PCC-20261114-59b70f; generated declarations are

`OrientationGradedNode.nodeRight.congr_simp`
`OrientationGradedNode.nodePolyRight.congr_simp`.

Fresh data is PCC-20261114-d3ab75, import-inclusive exit 0. All records remain unsigned [pcc:unverified].

Sources checked. KB unavailable as recorded in the journal; reviewed exact all-node numerator/content, selected-denominator, actual first-jet and row-refinement source proofs before extending them. Read the pinned primary D.4.3–7 and prior contraction-cost/precision/source-triage reviews. Standard algebra and refinement are reused without a novelty claim. All failed generic prototypes and raw outputs stay under the parent completion evidence directory. No native cryptographic, solver, timing, scaling, cloud or paid experiment ran. The old norm fixture remains unlaunched under its unchanged transport blocker.

The separate local cost5 and prior additive-FB source review are not part of these 42 statements or a new benchmark. No new measured data changes the canonical IC graphs/leaderboards. The graphic is a symbolic dependency map. PR4118 protected concurrent ledger histories still need legitimate canonical reconciliation; auto-merge stays off. No protected history or claim is rewritten.